Name ______________________________ Section ________

**MTH:2310 DISCRETE MATH**
**QUIZ 3**
**DUE: THURSDAY SEPT. 25**

*Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.*

**Handwriting:** *(4 points)*
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 2 points for arithmetic errors.
You will be penalized up to 2 points for disorganized or hard to read solutions.

**Problem 1.** *(10 points)*
Use a direct proof to prove the following claim.

*"For any integers $x, y$, and $z$, if $x^2$ divides $y$ and $y^3$ divides $z$, then $x^6$ divides $z$.*
**Def:** Given two integers $b$ and $a$, if $b \neq 0$ we say $b$ **divides** $a$ if there is an integer $c$ where $a = cb$ (or equivalently $\frac{a}{b} = c$).

Let there be three arbitrary integers $A, B, C$ where $A, B \neq 0$ such that $\frac{B}{A^2} = d$ and $\frac{C}{B^3} = g$ where $d, g \in \mathbb{Z}$

$B = A^2 d$, $B^3 = (A^2 d)^3 = A^6 d^3$, $C = A^6 d^3 g$ $\quad \frac{C}{A^6} = d^3 g$

As $\frac{C}{A^6} = d^3 g$, and $d^3 g \in \mathbb{Z}$, we conclude $A^6$ divides $C$

$\therefore$ for any integers $X, Y, Z$ if $X^2$ divides $Y$ and $Y^3$ divides $Z$, then $X^6$ divides $Z$

**Problem 2.** *(10 points)*
Use a proof by contraposition to prove the following claim.

*"If $x$ and $y$ are integers and $x^2(y^2 - 2y)$ is odd, then $x$ and $y$ are odd."*

$p(x,y) := x^2(y^2-2y)$ is odd $\rightarrow$ $q(x,y) := x \wedge y$ are odd

$\sim q(x,y) = x \vee y$ are even $\rightarrow \sim p(x,y) = x^2(y^2-2y)$ is even

Case 1: $x$ is even

Let $a, b, k$ be any even integer $a = 2k$

$4k^2(b^2-2b) = 4k^2b^2 - 8k^2b = 2(2k^2b^2 - 4k^2b)$

Case 2: $y$ is even

Let $a, b, k$ be any even integer $b = 2k$

$a^2(4k^2-4k) = 4k^2a^2 - 4ka^2 = 2(2k^2a^2 - 2ka^2)$

thus, we've proven that for all cases ( case 3 being covered as cases 1, 2 are agnostic for the parity of the non-forced variable)

given $x, y \in \mathbb{Z}$ $x^2(y^2-2y)$ is odd $\rightarrow x \wedge y$ are odd

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**Problem 3.** *(8 points)*
Use a direct proof, proof by contraposition, or proof by contradiction to prove the following claim.

*"If $n$ is any integer and $n^2$ is divisible by 4, then $n$ ~~is odd~~ even."*

by contraposition, if n is odd, then $n^2$ is not divisible by 4
Let n, k be any integer
$n = 2k+1$ $\quad n^2 = (2k+1)(2k+1) = 4k^2+4k+1$

$$\frac{4k^2+4k+1}{4} = k^2+k+\frac{1}{4} \qquad k^2+k+\frac{1}{4} \notin \mathbb{Z}$$

as $k^2+k+\frac{1}{4} \notin \mathbb{Z}$, using the axiom for parity and definition for divisibility, we prove by contraposition that if n is any integer, and $n^2$ is divisible by 4, then n is even

Prove (n is even) → ($n^2$ is Divisible by 4)

Let $a, b \in \mathbb{Z}$ such that $a = 2b$

$$(2b)^2 = 4b^2 \qquad \frac{4b^2}{4} = b^2$$

as $b^2 \in \mathbb{Z}$, there exists an integer such that $a^2$ is Divisible by, thus proving

(n is even) → ($n^2$ is Divisible by 4)

**Problem 4.** *(10 points)*
Use a proof by contradiction to prove the following claim.

*"Given any integers $x, y$, and $z$, if $x^2 + y^2 = z^2$ then $x$ or $y$ is even."*

below is the proof using modulus. that's not the proof to be graded, but I spent a lot of time on it so it holds sentimental value :)

by contradiction: $x^2+y^2=z^2 \to x \wedge y$ are odd

Let $a,b \in \mathbb{Z}$ such that $a=2c$, $b=2d+1$ $c,d \in \mathbb{Z}$

$a^2 = 4c^2$ $\quad b^2=4d^2+4d+1$

$4c^2 \bmod 4 = 0$ $\quad (4d^2+4d+1) \bmod 4 = (4d^2 \bmod 4) + (4d \bmod 4) + (1 \bmod 4) = 1$

$C_0$ · Thus the modulos 4 of a squared even number $=0$, and

$C_1$ · the modulos 4 of a squared odd number $=1$

Let $a,b,c \in \mathbb{Z}$ such that $a=2d+1$, $b=2e+1$ $d,e \in \mathbb{Z}$

$(2d+1)^2+(2e+1)^2 = c^2 \equiv 4d^2+4d+4e^2+4e+2 = c^2$ $\quad c^2 \bmod 4 = (4d^2+4d+4e^2+4e+2) \bmod 4 = 2$

As $c^2 \bmod 4 = 2$, $C_1$ means $c \notin \mathbb{Z}$

we have proven that given any integers $x, y, z$ if $x^2+y^2=z^2$ then $x$ or $y$ is even

below is the proof to be graded

Suppose for contradiction $x^2+y^2=z^2 \to x \wedge y$ are odd

Let $a,b,c,d,g \in \mathbb{Z}$ such that $a=2c+1$, $b=2d+1$

$(2c+1)^2+(2d+1)^2 = g^2 \equiv 4c^2+4c+4d^2+4d+2 = g^2$

$$\frac{4c^2+4c+4d^2+4d+2}{4} = c^2+c+d^2+d+\frac{1}{2}$$

thus, $g^2$ is not Divisible by 4, and therefore $g$ cannot be even, which means $g$ must be odd.

Let $j, k \in \mathbb{Z}$ $\quad j = 2k+1$

$$(2k+1)^2 = 4k^2+4k+1 = 2(2k^2+2k)+1$$

thus an odd number squared, is an odd number.

$$g^2 = 4c^2+4c+4d^2+4d+2 = 2(2c^2+2c+2d^2+2d+1)$$

thus as $g^2$ is an even number, as $g$ is odd, and the square of an odd number must be odd, we've proven

$$(x^2+y^2=z^2) \to (x \text{ or } y \text{ must be even})$$